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Subject : DSA

Unit 2, S5, SLO1 - Singly Linked List – Traversal and Searching

Algorithm for traversing a linked list

Step 1: [INITIALIZE] SET PTR = _____

Step 2: Repeat Steps 3 and 4 while _____

Step 3: Apply Process to _____

Step 4: SET PTR = _____

[END OF LOOP]

Step 5: EXIT

Answer: Step 1: SET PTR = START

Step 2: Repeat Steps 3 and 4 while PTR != NULL

Step 3: Apply Process to PTR -> DATA

Step 4: SET PTR = PTR -> NEXT

Step 5: EXIT

Algorithm to print the number of nodes in a linked list

Step 1: [INITIALIZE] SET COUNT = 0

Step 2: [INITIALIZE] SET _____

Step 3: Repeat Steps 4 and 5 while _____

Step 4: SET COUNT = _____

Step 5: SET PTR = _____

[END OF LOOP]

Step 6: Write _____

Step 7: EXIT

Answer: Step 2: SET PTR = START

Step 3: Repeat Steps 4 and 5 while PTR != NULL

Step 4: SET COUNT = COUNT + 1

Step 5: SET PTR = PTR -> NEXT

Step 6: Write COUNT

Step 7: EXIT

Algorithm to search a linked list

Step 1: [INITIALIZE] SET PTR = _____

Step 2: Repeat Step 3 while PTR != _____

Step 3: IF VAL = _____

SET POS = _____

Go To Step 5

ELSE

SET PTR = _____

[END OF IF]

[END OF LOOP]

Step 4: SET POS = _____

Step 5: EXIT

Answer: Step 1: SET PTR = START

Step 2: Repeat Step 3 while PTR != NULL

Step 3: IF VAL = PTR -> DATA

SET POS = PTR

Go To Step 5

ELSE

SET PTR = PTR -> NEXT

Step 4: SET POS = NULL (or -1)

Step 5: EXIT